



HWA CHONG INSTITUTION
JC2 Preliminary Examination
Higher 1

CANDIDATE NAME

CT GROUP

CENTRE NUMBER

INDEX NUMBER

BIOLOGY

8875 / 02

Paper 2 Core Paper

3 September 2013

Additional Materials: Writing Paper

2 hours

INSTRUCTIONS TO CANDIDATES

Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page.

SECTION A

This section contains **four** structured questions. Answer **all** questions.

Write your answers on the lines / in the spaces provided.

SECTION B

This section contains **two** questions. Answer any **one** question.

Your answers must be in continuous prose, where appropriate.

Write your answers on the writing paper provided.

BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.

A **NIL RETURN** is required.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You may use a calculator.

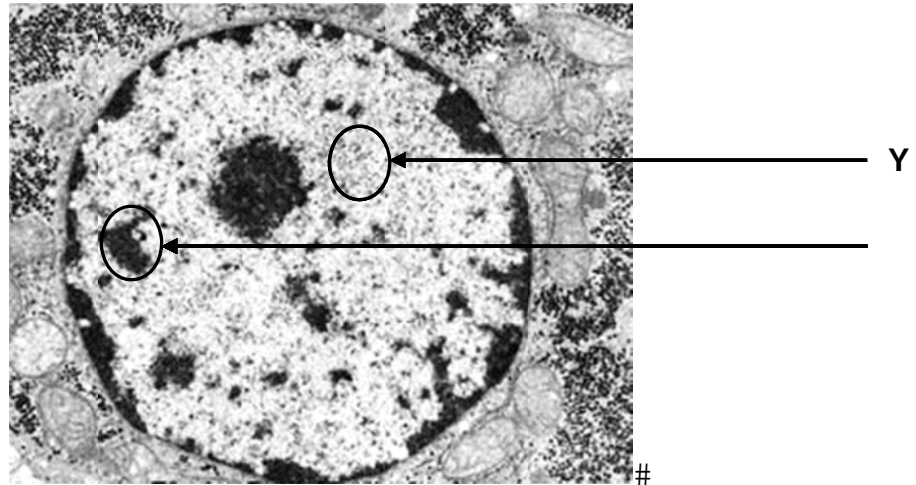
You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
Question	Marks
1	/ 12
2	/ 7
3	/ 11
4	/ 10
5 / 6	/ 20
Total	/ 60

SECTION A: STRUCTURED QUESTIONS

QUESTION 1

Fig. 1.1 shows an electron micrograph of part of a secretory cell obtained from a mouse.



#

Fig. 1.1

- (a) (i)** Name structures **X** and **Y**. [2]

X: _____

Y: _____

- (ii)** Account for the difference in structures **X** and **Y**. [2]

Fig. 1.2 shows part of a DNA molecule.

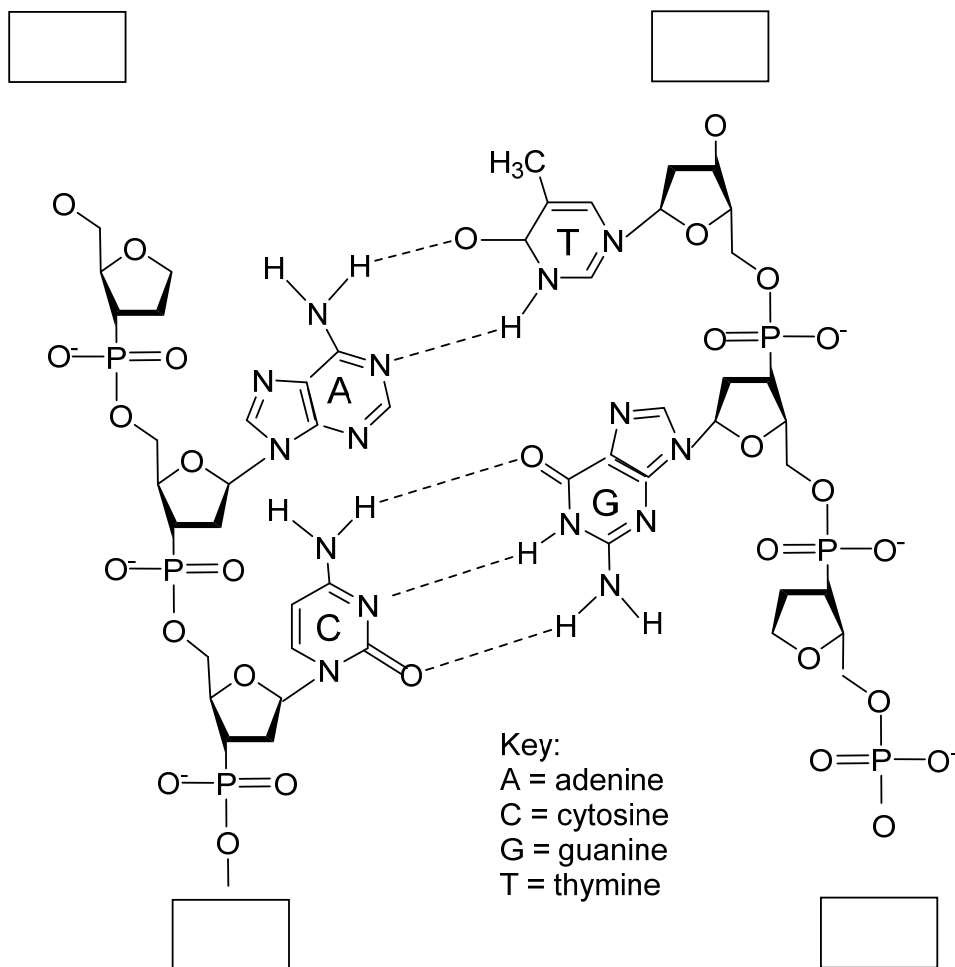


Fig. 1.2

- (b) (i) Complete Fig. 1.2 by indicating the polarity of the DNA molecule in the boxes provided. [1]
- (ii) State the importance of hydrogen bonds in DNA structure. [2]

In the 1950s, Erwin Chargaff determined the relative quantities of the four bases in DNA in different organisms. His results provided important evidence for the model of DNA proposed by James Watson and Francis Crick in 1953. Some of Chargaff's data is shown in Table 1.1.

Table 1.1

organism	percentage of adenine	percentage of thymine	percentage of guanine	percentage of cytosine
<i>Escherichia coli</i> (bacterium)	24.7	23.6	26.0	25.7
yeast	31.3	32.9	18.7	17.1
wheat	27.3	27.1	22.7	22.8
octopus	33.2	31.6	17.6	17.6
sea urchin	32.8	32.1	17.7	17.3
chicken	28.0	28.4	22.0	21.6
human	29.3	30.0	20.7	20.0

(c) Explain how the data in Table 1.1 helps to confirm the arrangement of bases in DNA. [3]

Table 1.2 shows Chargaff's data for a virus.

Table 1.2

organism	percentage of adenine	percentage of thymine	percentage of guanine	percentage of cytosine
virus	24.0	31.2	23.3	21.5

- (d) (i)** State how the results for the virus differ from those for all the organisms given in Table 1.1. [1]

- (ii)** Give a reason for your answer to **(d) (i)**. [1]

[Total: 12]

QUESTION 2

A genetic disorder is an illness caused by one or more abnormalities in the genome. Albinism and red-green colour blindness are recessive disorders caused by 2 different genes found on chromosome 11 and chromosome X respectively.

- (a)** Describe how an individual can inherit albinism from his parents. [2]

An albino man who has normal red-green colour vision marries a woman who has normal skin pigment and normal red-green colour vision. She is heterozygous at both gene loci.

- (b) Using the symbols provided, draw a genetic diagram in the space below to show the cross between the couple. [5]

Let **N** represent the dominant allele for normal skin pigment
n represent the recessive allele for no skin pigment
B represent the dominant allele for normal red-green colour vision
b represent the recessive allele for red-green colour blindness

[Total: 7]

QUESTION 3

In a study on the role of proton transfer through the ATP synthase complex, a plant was subjected to flashes of light (indicated by the arrow) and the concentration of ATP synthesised was subsequently recorded as shown in Fig. 3.1. The experiment was also conducted in the presence of venturicidin, a potent inhibitor of proton flow in ATP synthase complex.

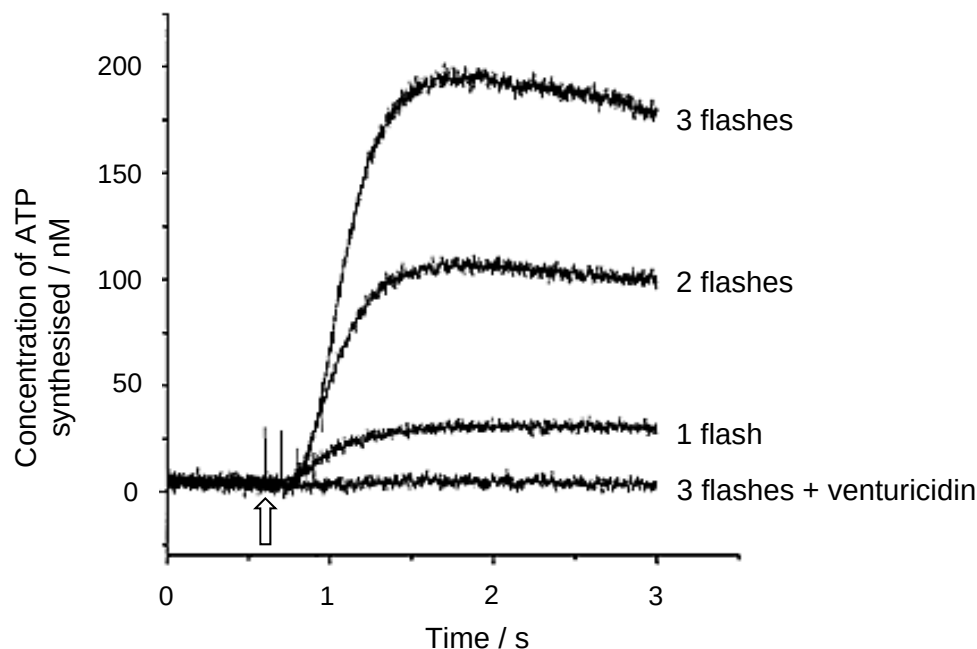


Fig. 3.1

- (a) (i) Describe the effect of flashing light on ATP synthesis in the absence of venturicidin. [2]

(ii) Explain your answer to (a) (i).

[4]

(b) Describe and explain the effect of venturicidin on the plant and its ability to produce sugars.

[4]

(c) Distinguish the role of ATP synthesis in mitochondria and chloroplasts.

[1]

[Total: 11]

QUESTION 4

Bovine somatotrophin is a peptide hormone produced by cows' pituitary glands in minute amount to regulate metabolic processes. With the use of recombinant DNA technology, it is now possible to large-scale produce recombinant bovine somatotrophin (rBST) in bacteria. rBST is injected into cow to increase milk production.

In the creation of rBST, scientist inserted the gene coding for bovine somatotrophin in a plasmid as shown in Fig. 4.1.

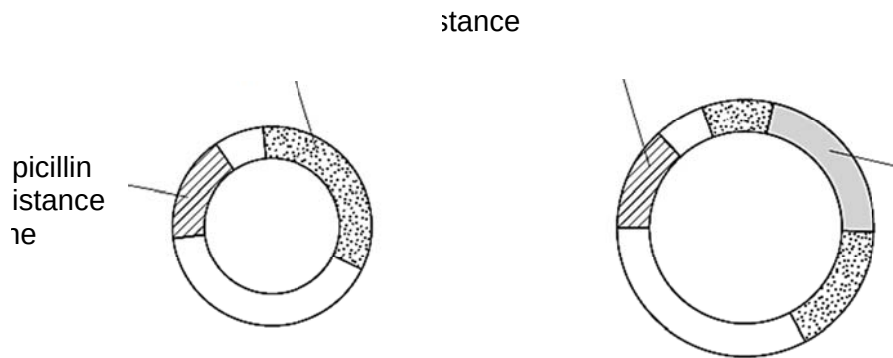


Fig. 4.1

- (a) Describe how the gene coding for bovine somatotrophin is inserted into the plasmid. [3]

The recombinant plasmids were then transformed into bacteria. Replica plating was used to identify the bacteria with the gene coding for bovine somatotrophin. Fig. 4.2 shows the bacterial colonies that grew on two replica plates.

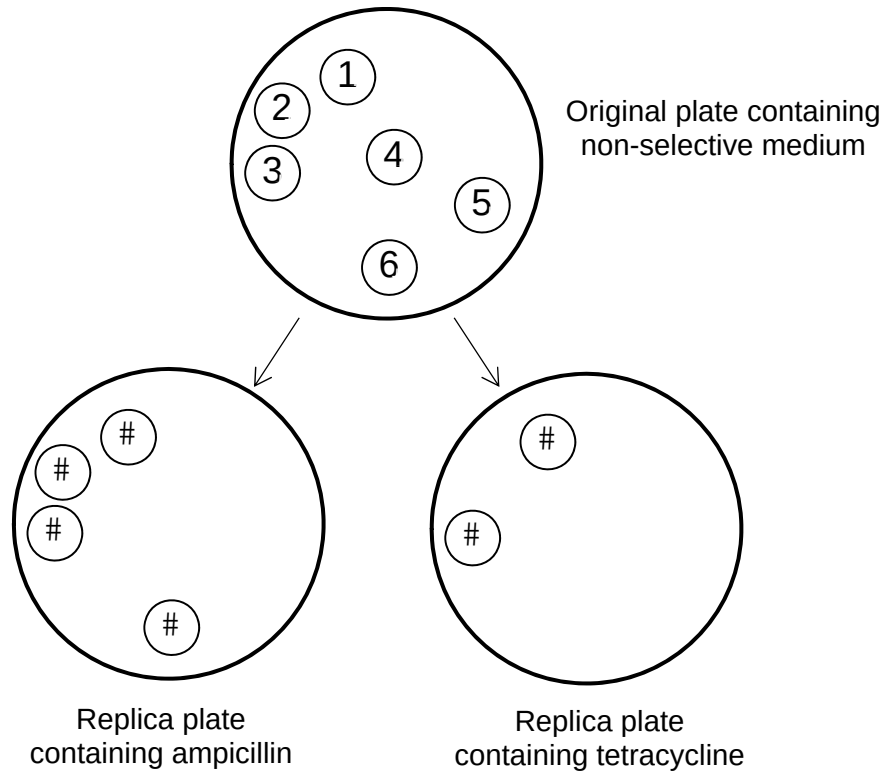


Fig. 4.2

(b) Account for the results shown in the replica plates.

[3]

A pharmaceutical company wanted to increase the yield of milk produced. Genetically modified cows were created by inserting the gene coding for bovine somatotrophin into adult stem cells in their mammary glands. They carried out an experiment to compare the average volume of milk produced yearly by 1000 cows of different age groups with injected rBST and that of genetically modified cows. The results of the comparison are shown in Fig. 4.3.

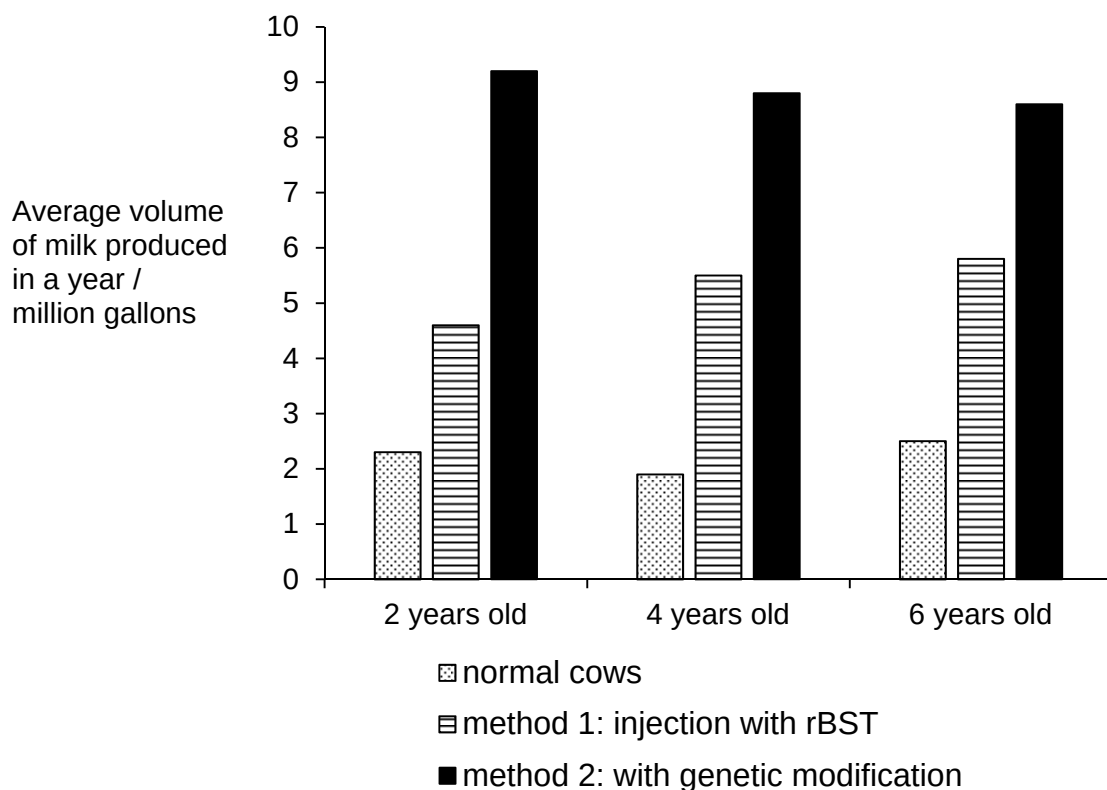


Fig. 4.3

- (c) Discuss which method should be used by the company to increase milk production in cows. [4]

[Total: 10]

--- END OF SECTION A ---

SECTION B: FREE RESPONSE QUESTION

Answer **ONE** question.

Your answers must be in continuous prose, where appropriate.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.

A **NIL RETURN** is required.

QUESTION 5

- (a) Explain how the structure of haemoglobin and glycogen are related to their function. [8]
- (b) Outline the mode of action of restriction enzymes. [6]
- (c) Describe the effect of inhibitors on the rate of enzyme activity. [6]

QUESTION 6

- (a) Describe the normal functions of embryonic stem cells and blood stem cells in a living organism. [8]
- (b) Discuss how meiosis and random fertilisation can lead to genetic variation. [6]
- (c) Explain, with a named example, how environmental factors act as forces of natural selection. [6]

--- END OF SECTION B ---

--- End of Paper ---